

ECE 105: Introduction to Electrical Engineering

Lecture 21

Optical Communications

Yasser Khan

Rehan Kapadia

How have we done optical communications in the past?

What kind of systems have people used in the past to do optical communications?



Smoke Signals



Signal Lamp



Heliograph

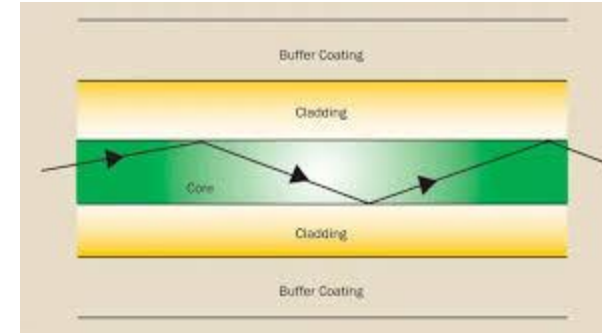
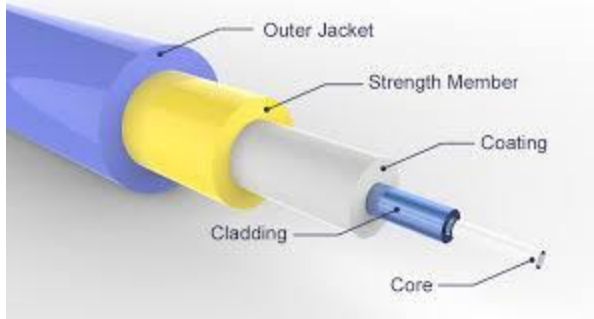


Fiber Optics

How fast could we communicate using smoke signals?

- We could have a short or long pulse to represent a 1 or 0.
- Let's assume 0.5 seconds for short and 1.5 seconds for long pulse
- The gap between the pulses needs to be at least 0.5 seconds for it to be seen
- That means that we would need $0.5+0.5$ to send out a short pulse
- We would need $1.5+0.5$ to send out a long pulse
- Average time to send a bit = $(\text{Short} + \text{Long})/2 = 1.5 \text{ s}$
- Our bitrate is $1 \text{ bit}/1.5\text{s} = 0.66 \text{ bps}$

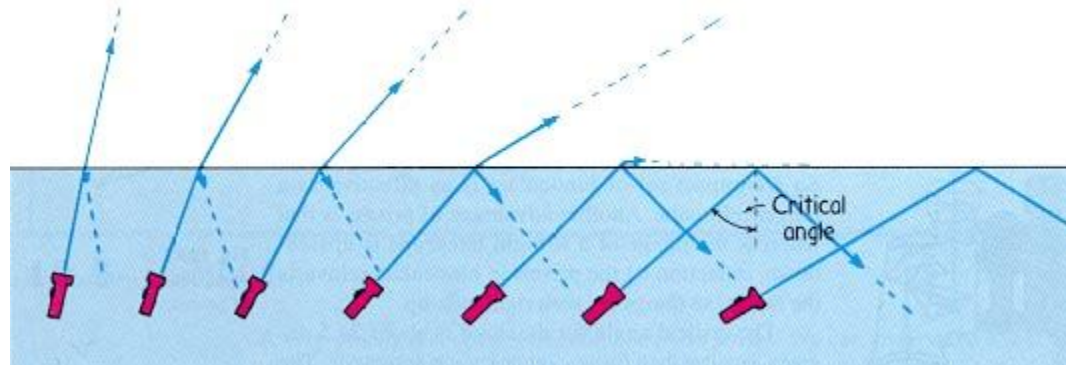
Optical Fiber



Optical fibers guide light from one point to another confined inside a fiber core.

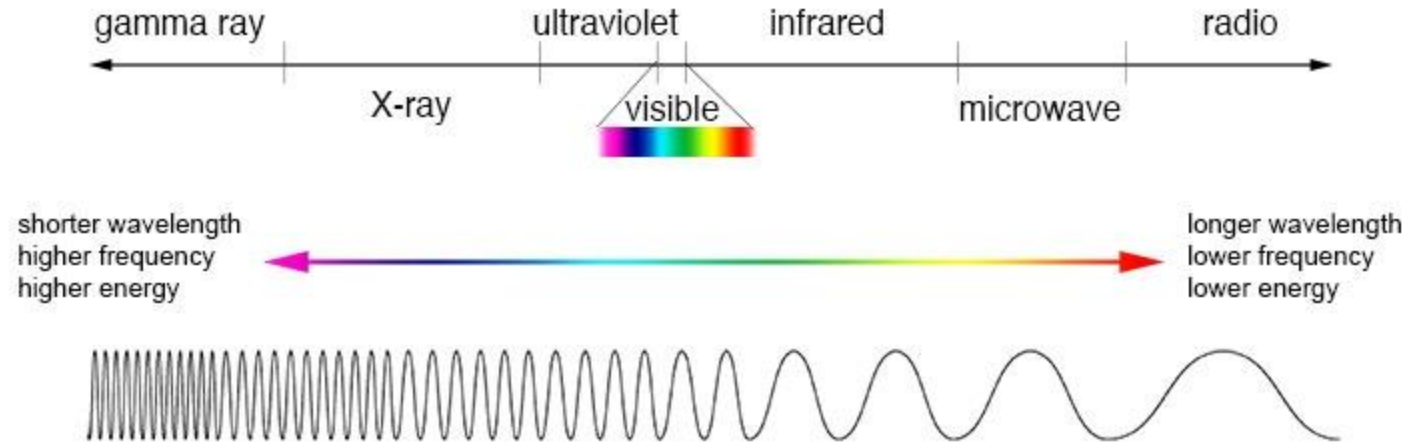
These work based on the principle of total internal reflection. Depending on the angle that light hits the fiber/cladding interface, it can be fully reflected

Total internal reflection



When light moves from a medium of high refractive index to a medium of low refractive index, or vice versa, there is a change in the angle of propagation of light.

General properties of light



The wavelength of light represents the spatial dimension over which it is periodic

Speed of light in different mediums

There is a concept called the refractive index, which is:

$$n = c/v$$

Where

n: refractive index

c: speed of light

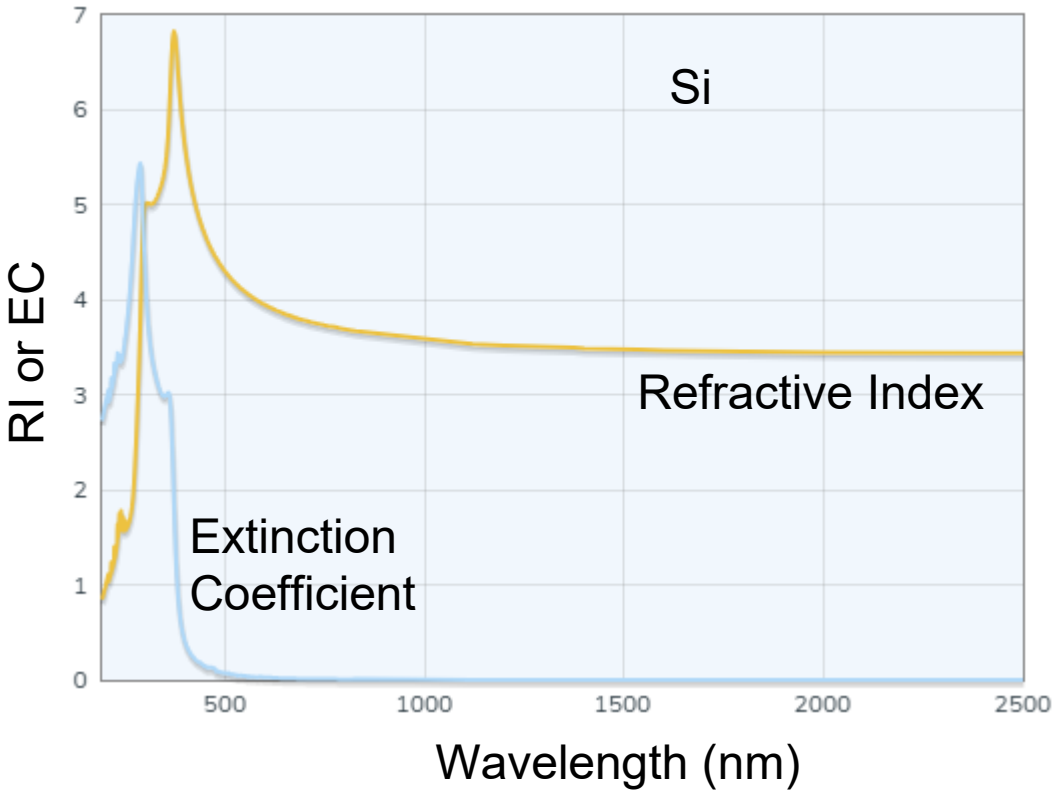
v: speed of light in the medium

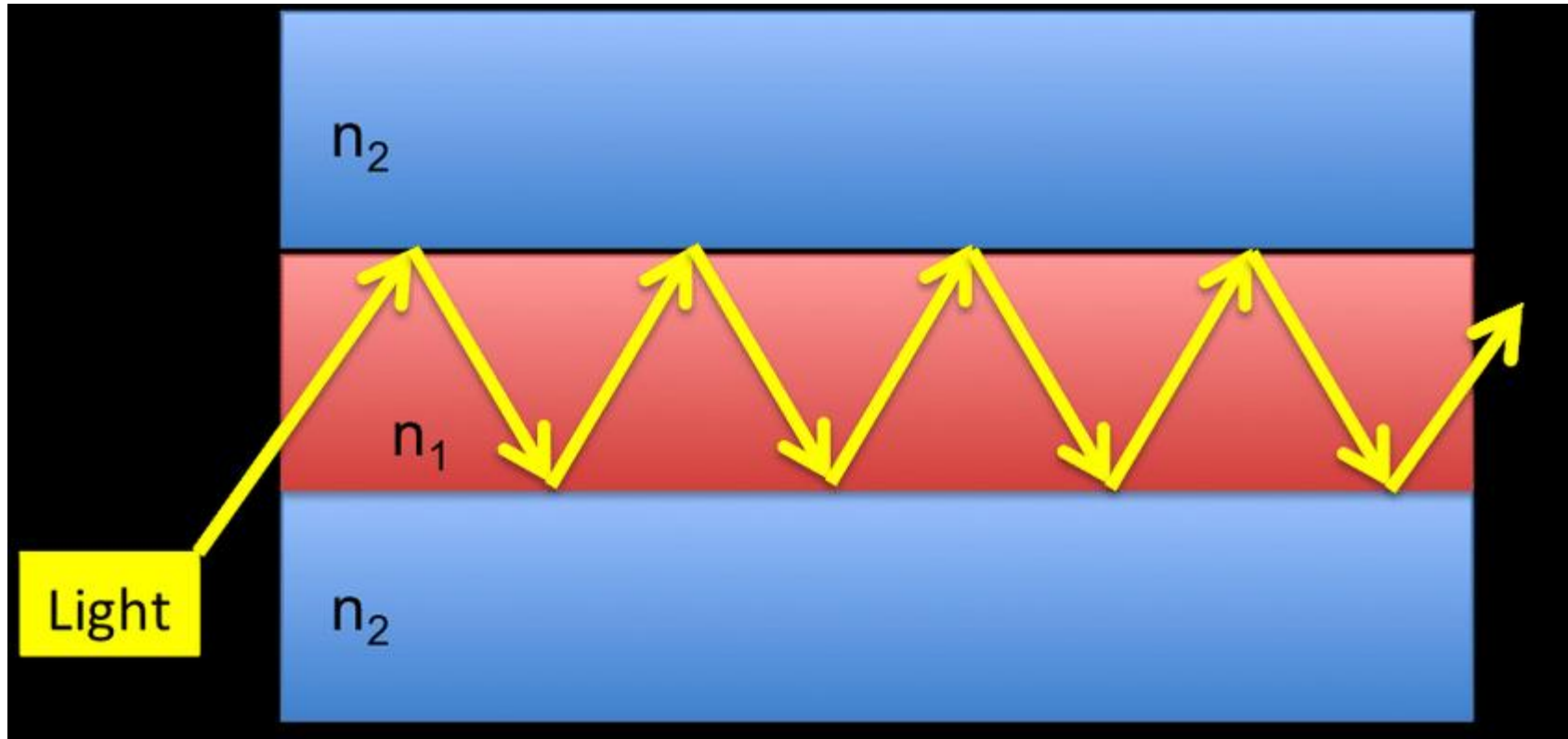
This emerges due to the oscillations of electrons in a material creating dipoles that interact with electromagnetic waves and slow them down

Refractive index of different materials

Material	<i>n</i>
Vacuum	1
Gases at 0 °C and 1 atm	
Air	1.000 293
Helium	1.000 036
Hydrogen	1.000 132
Carbon dioxide	1.000 45
Liquids at 20 °C	
Water	1.333
Ethanol	1.36
Olive oil	1.47
Solids	
Ice	1.31
Fused silica (quartz)	1.46 ^[11]
PMMA (acrylic, plexiglas, lucite, perspex)	1.49
Window glass	1.52 ^[12]
Polycarbonate (Lexan™)	1.58 ^[13]
Flint glass (typical)	1.69
Sapphire	1.77 ^[14]
Cubic zirconia	2.15
Diamond	2.417
Moissanite	2.65

The refractive index is a function of material AND wavelength of the electromagnetic radiation





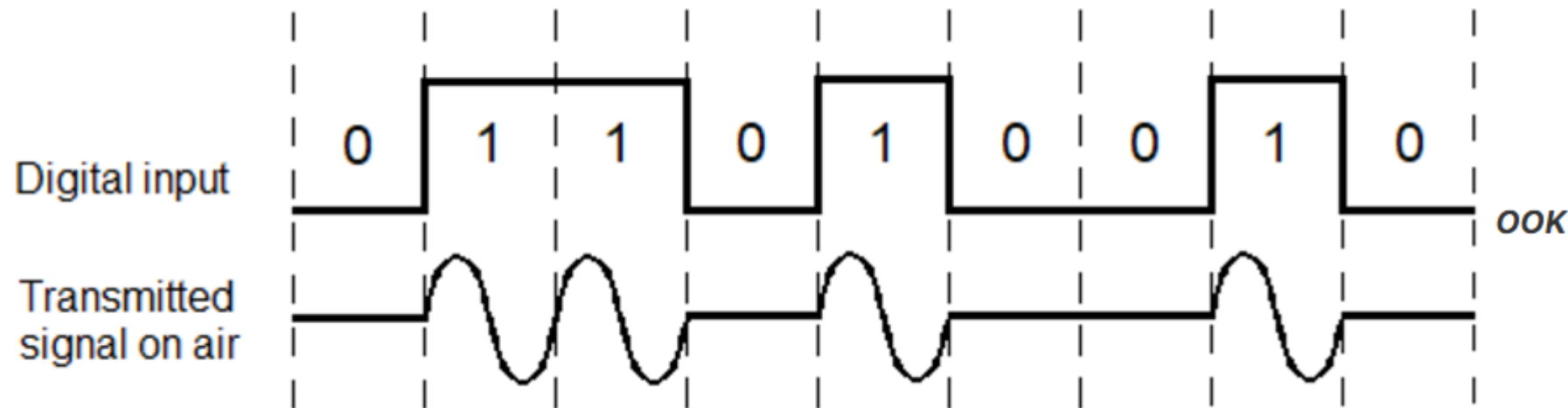
If light is incident on an interface at an angle beyond the critical angle, then it is reflected back. It will also be incident on the lower surface at an angle that enables total internal reflection. This allows us to create a waveguide, where light will just propagate in a confined guide as long as possible.

Optical Fiber Transmission Speeds

Year	Organization	Aggregate speed	Bandwidth	Spectral efficiency, (bit/s)/Hz	WDM channels	Per-channel speed	Distance
2009	Alcatel-Lucent ^[39]	15.5 Tbit/s			155	100 Gbit/s	7000 km
2010	NTT ^[40]	69.1 Tbit/s			432	171 Gbit/s	240 km
2011	NEC ^[41]	101.7 Tbit/s			370	273 Gbit/s	165 km
2011	KIT ^{[42][43]}	26 Tbit/s			336 ^[A]	77 Gbit/s	50 km
2016	BT & Huawei ^[44]	5.6 Tbit/s			28	200 Gbit/s	~140 km?
2016	Nokia Bell Labs, Deutsche Telekom & Technical University of Munich ^{[45][46]}	1 Tbit/s		5–6.75	4	250 Gbit/s	419–951 km
2016	Nokia-Alcatel-Lucent ^[47]	65 Tbit/s					6600 km
2017	BT & Huawei ^[48]	11.2 Tbit/s		6.25	28	400 Gbit/s	250 km
2020	RMIT, Monash & Swinburne Universities ^{[49][50]}	39.0–40.1 Tbit/s	~4 THz	10.4 (10.1–10.4)	160 ^[A]	244 Gbit/s	76.6 km
2020	UCL ^[51]	178.08 Tbit/s	16.83 THz	10.8	660 (S, C, L bands)	270 Gbit/s	40 km
2023	NICT ^[52]	301 Tbit/s	27.8 THz	10.8	1097 (E, S, C, L bands)	250–300 Gbit/s	50–150 km
2024	NICT ^[53]	402 Tbit/s	37.6 THz	10.7	1505 (O, E, S, C, L, U bands)	170–320 Gbit/s	50 km

Intensity Modulation: Where you modulate the intensity of the light, and have predetermined values for each intensity of light.

On-off keying is where you have two values that represent 1 and 0, allowing you to transmit binary information



Let's talk about a real world system



Let's imagine a system where we would like to send information from one place to another using light.

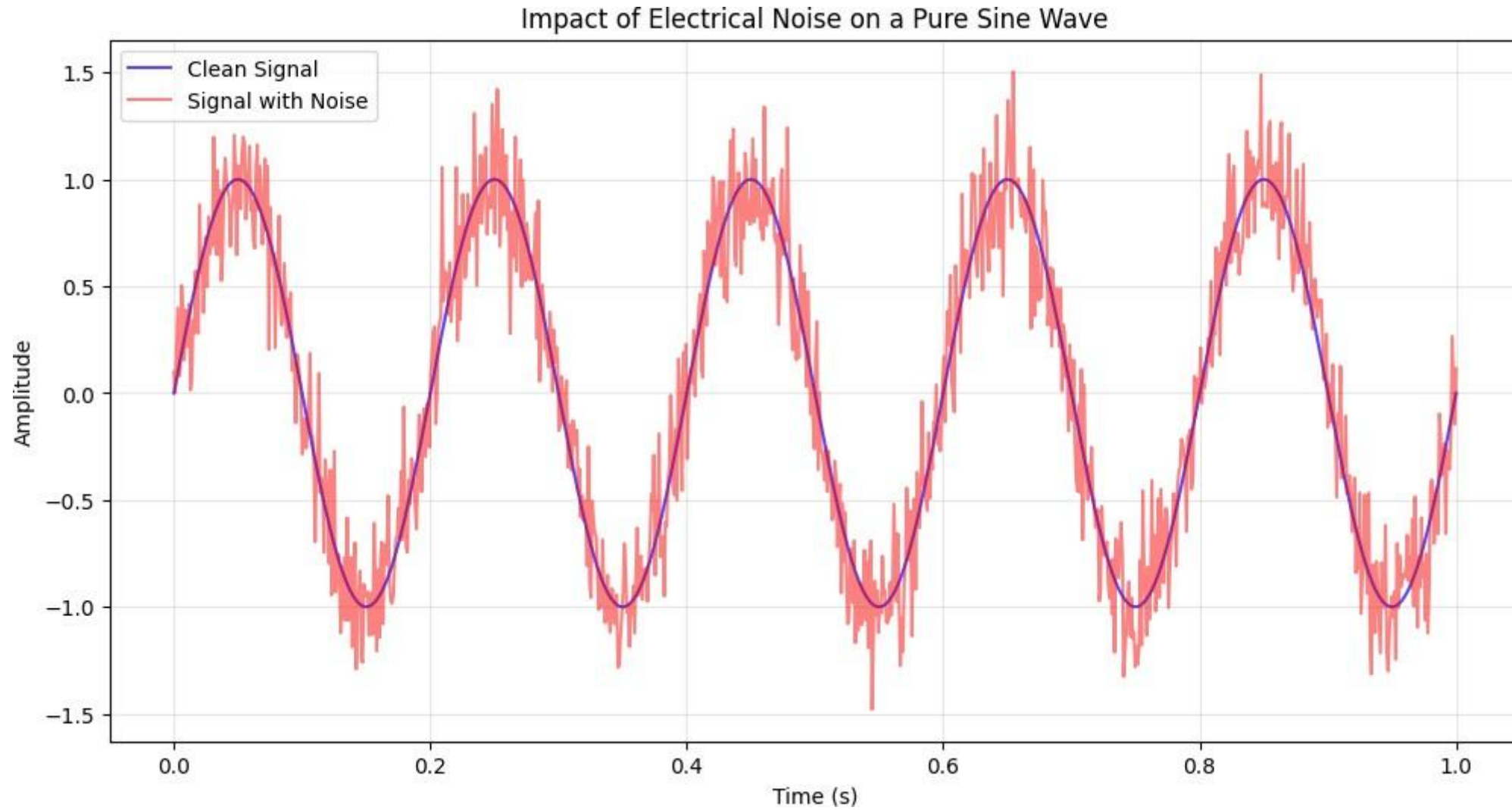
Seems easy, right? We just flash an LED, detect those flashes, and then we are done.

But how much data could we send with such a signal?

What components would we need?

How can we accomplish this?

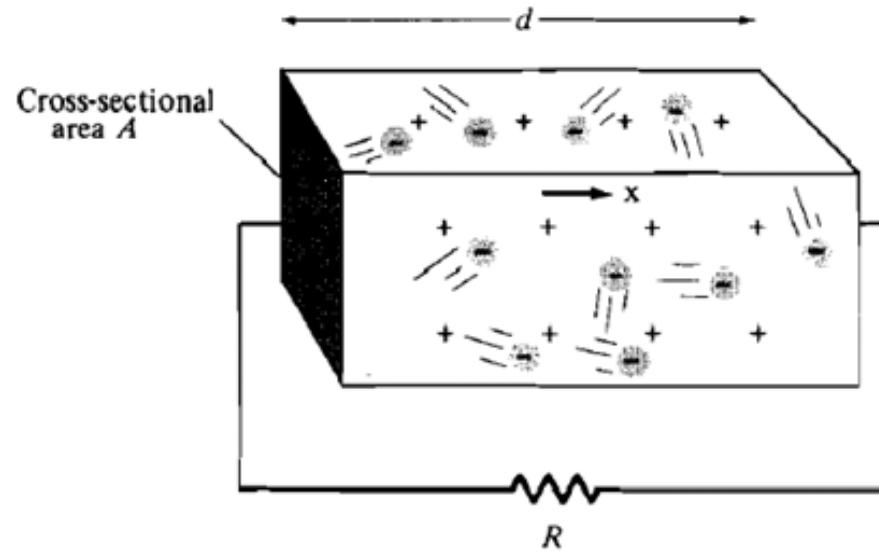
What is noise?



- Noise refers to anything that you don't want in your signal
- It can have many different sources and many different resulting affects on your system
- At the component level, here are a few fundamental mechanisms of noise:
 - Thermal Noise (Johnson-Nyquist): Arises from the random motion of charge carriers in resistive materials.
 - Flicker Noise: Generally related to defects and impurities
 - Shot Noise: Arises from the discrete nature of charge carriers

Thermal Noise

- **Fluctuation in the voltage across a dissipative circuit element (resistor)**
- **Caused by thermal motion of charged carriers**



$$v_{n,rms} = \sqrt{4kTB}$$

Here's what each symbol means:

- $v_{n,rms}$
RMS noise voltage across the resistor (what you ultimately care about).
- k – Boltzmann constant
 $k \approx 1.38 \times 10^{-23} \text{ J/K}$
Sets the scale of thermal energy per degree Kelvin.
- T – Absolute temperature
In Kelvin (e.g., room temp $\approx 300 \text{ K}$).
Higher $T \Rightarrow$ more thermal agitation $\Rightarrow$ more noise.
- R – Resistance
In ohms (Ω).
Larger $R \Rightarrow$ larger thermal voltage noise.
- B – Noise bandwidth
In Hz. This is the effective frequency range over which you're measuring (e.g., 100 kHz in your problem).
Larger $B \Rightarrow$ more integrated noise power.

Shot Noise

Shot noise is caused by the random arrival of charge carriers (e.g., electrons or holes) at a potential barrier or through a quantum tunneling process. Unlike thermal noise, which depends on temperature, shot noise arises solely due to the quantization of charge.

Key Characteristics:

- It is independent of temperature.
- It is proportional to the average current and depends on the bandwidth of observation.
- It is most commonly observed in **semiconductor junctions**, **vacuum diodes**, and **photodetectors**.

$$i_{\text{shot,rms}} = \sqrt{2qIB}$$

Here's what each symbol means and why it matters:

- $i_{\text{shot,rms}}$
RMS **shot noise current** (A). This is the fluctuating part of the current due to the random arrival of discrete charges (electrons).
- q – *Electron charge*
 $q \approx 1.6 \times 10^{-19}$ C.
This is what makes the current "granular" — current is made of individual charges, not a continuous fluid.
- I – *Average (DC) current*
In amperes (A).
For a photodiode, this is typically the **photocurrent** (plus possibly dark current).
Bigger $I \Rightarrow$ more electrons per second $\Rightarrow$ higher shot noise.
- B – *Noise bandwidth*
In Hz. This is the effective frequency range of your measurement (e.g., 100 kHz).
Larger $B \Rightarrow$ more noise power integrated $\Rightarrow$ higher RMS noise.

- **Proportional to Current:**

Shot noise increases linearly with the current I . Systems with higher currents generate more shot noise.

- **Statistical Nature:**

Shot noise is inherently random and follows a Poisson distribution because it is governed by the discrete and independent arrival of charge carriers.

- **Independence from Temperature:**

Unlike thermal noise, shot noise does not depend on the ambient temperature, making it a fundamental noise source.

- **Quantum Nature:**

Shot noise is a direct consequence of the quantized nature of charge, highlighting its quantum mechanical origin.

Definition:

Flicker noise, also known as $1/f$ noise, is a type of electrical noise that is dominant at low frequencies. It is caused by imperfections and defects in materials or processes and is observed in electronic devices like resistors, transistors, and semiconductors.

Key Characteristics:

•Frequency Dependence:

The power spectral density (PSD) decreases as the frequency increases, following a $1/f^\alpha$ relationship where α is typically between 0.8 and 1.2.

•Low-Frequency Dominance:

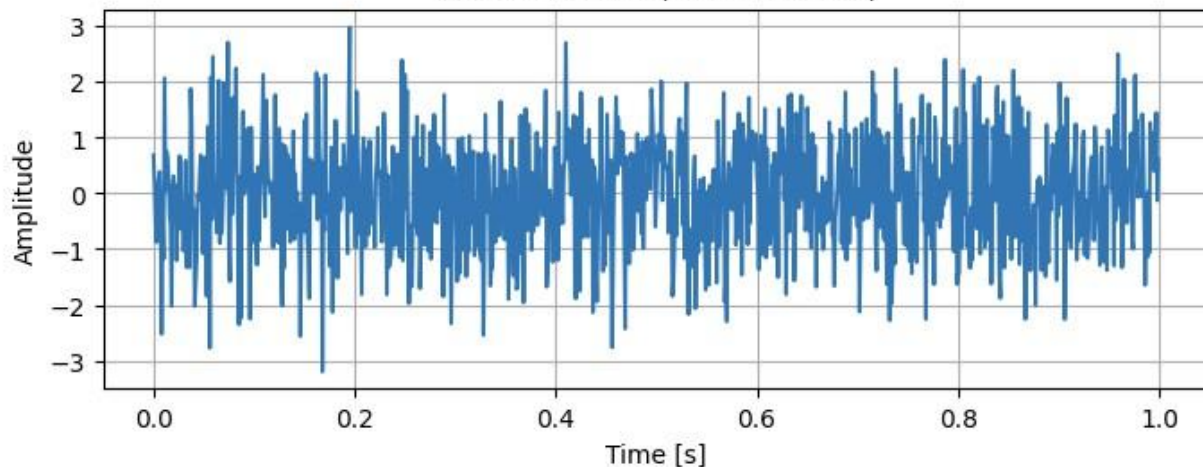
Flicker noise is most prominent at lower frequencies and becomes negligible at higher frequencies.

•Material-Dependent:

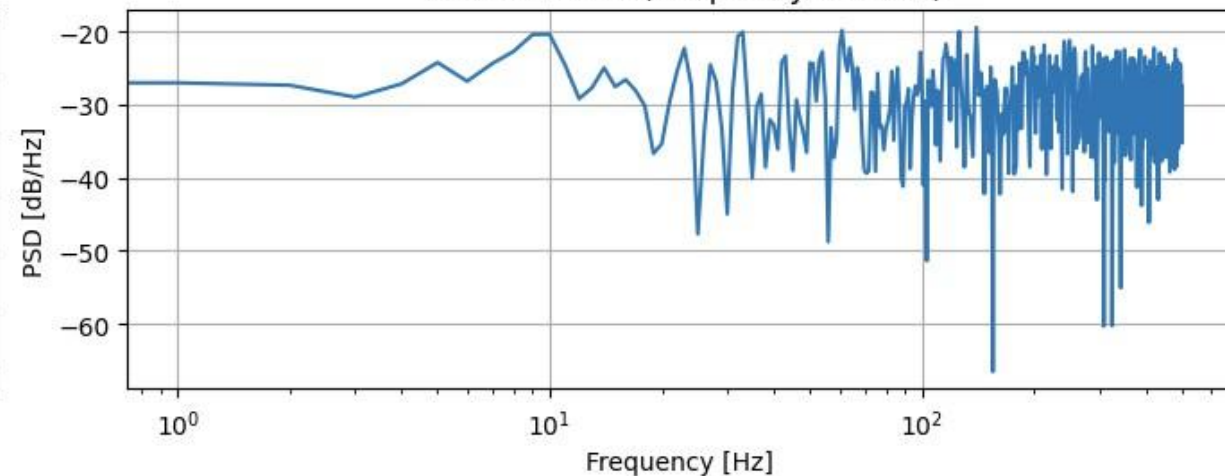
It arises from defects, impurities, and surface traps in materials.

Time and Frequency Domain Visualizations

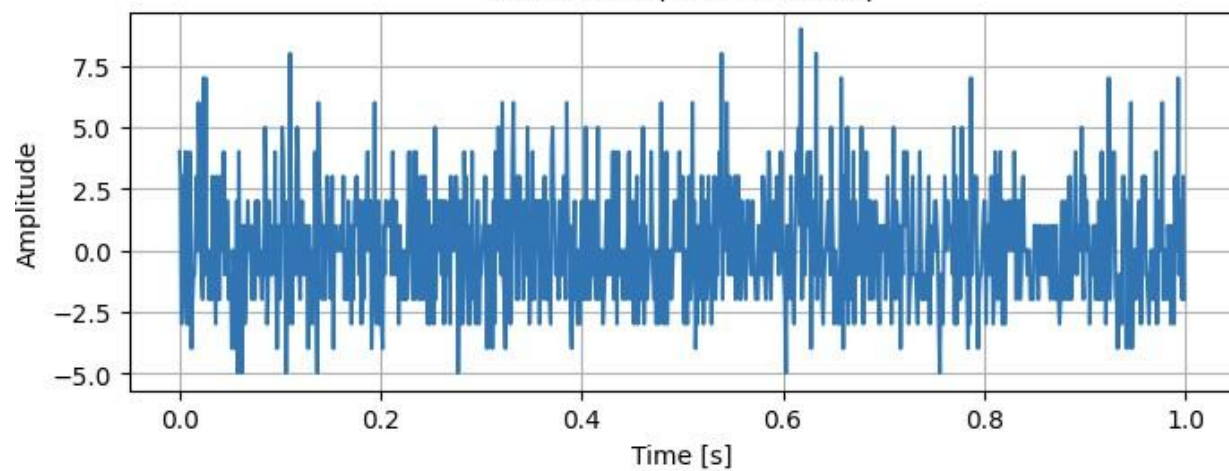
Thermal Noise (Time Domain)



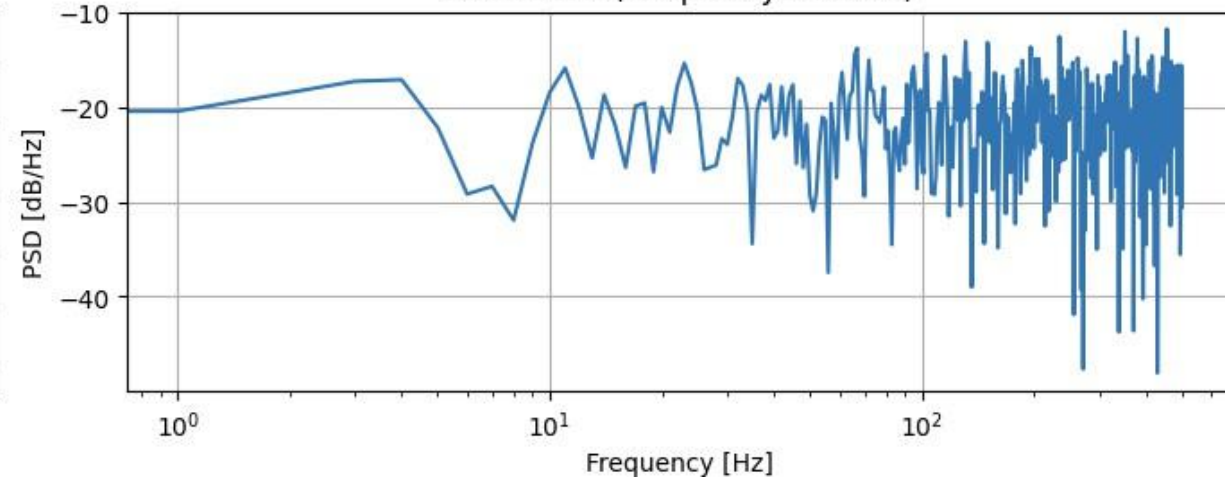
Thermal Noise (Frequency Domain)



Shot Noise (Time Domain)



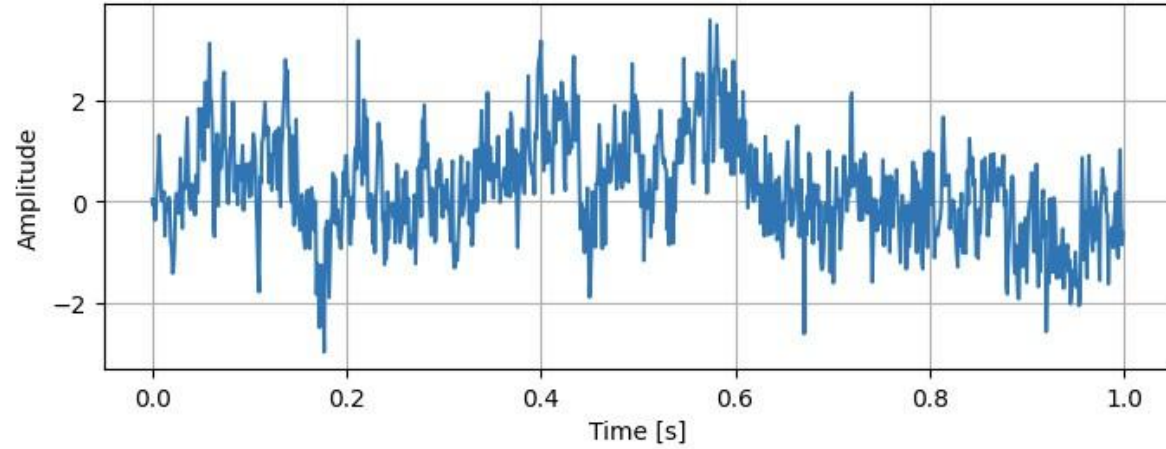
Shot Noise (Frequency Domain)



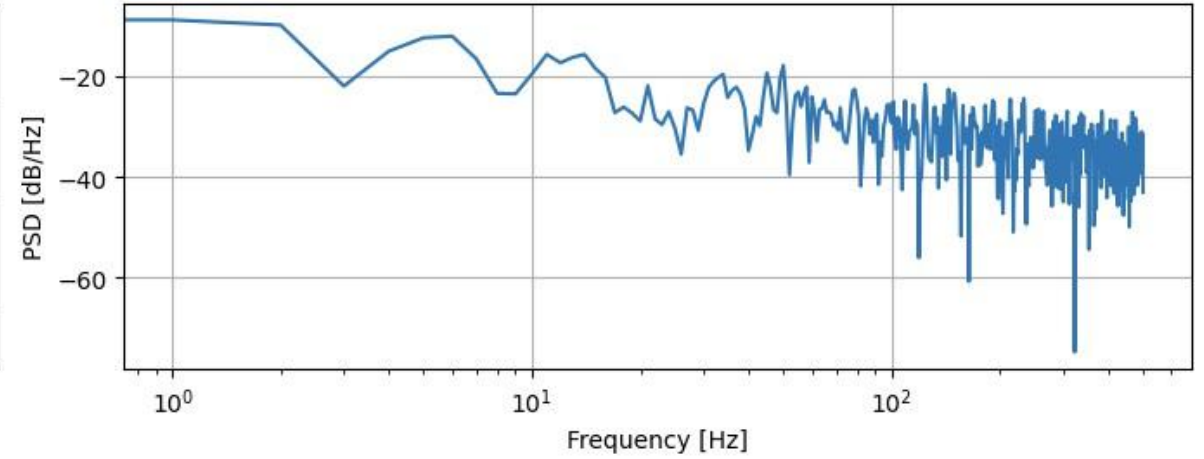
Time and Frequency Domain Visualizations

apse Output

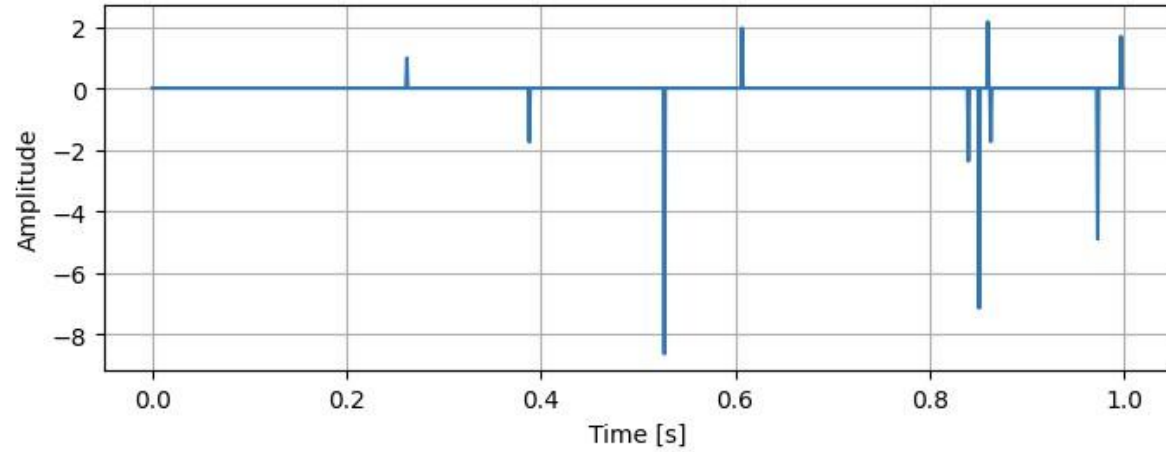
Flicker Noise (Time Domain)



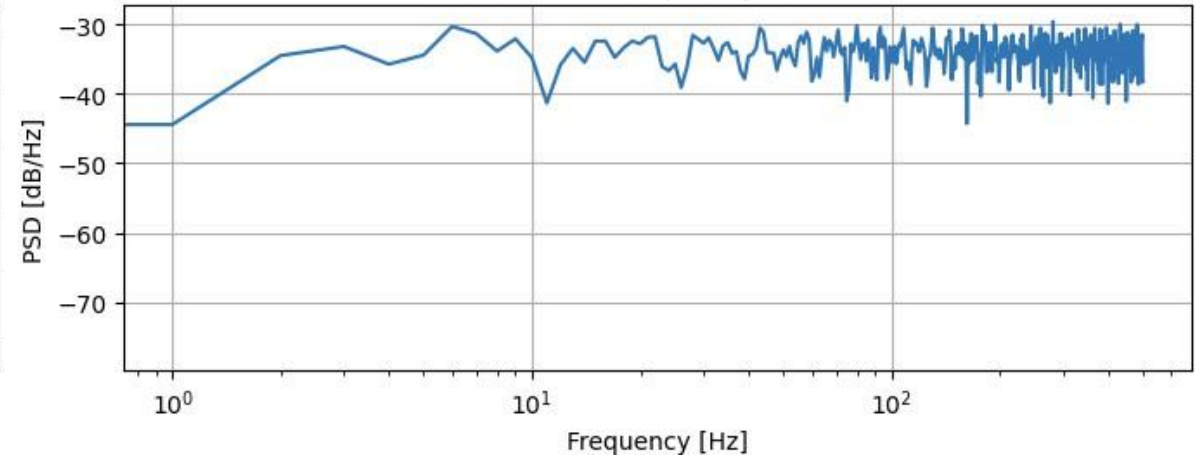
Flicker Noise (Frequency Domain)



Burst Noise (Time Domain)



Burst Noise (Frequency Domain)



How do different noise sources add?

When multiple uncorrelated noise sources contribute to the total noise in a system, the **total RMS noise** is calculated using the **root-sum-square (RSS)** method:

$$V_{n_{total}} = \sqrt{V_{n_1}^2 + V_{n_2}^2 + \cdots + V_{n_n}^2}$$

where:

- $V_{n_1}, V_{n_2}, \dots, V_{n_n}$ are the RMS noise values of individual noise sources.

This summation is based on the principle that the **variances** of uncorrelated random variables can be added together to get the total variance, and then the RMS value is the square root of this variance.

Why Do We Use Quadrature Summation?

- Noise sources are typically random processes that vary both in amplitude and phase.
- Since the different noise sources are **uncorrelated**, they do not reinforce each other directly as coherent signals would.
- Therefore, we must use the **root-sum-square** approach, which combines their powers rather than their amplitudes.

Power Summation Approach (Variance Approach)

The quadrature summation of noise RMS values can be understood from the perspective of **noise power**. For a random signal, the power (P) is proportional to the **square of the RMS value**. Thus, the total noise power from multiple uncorrelated sources is:

$$P_{n_{total}} = P_{n_1} + P_{n_2} + \cdots + P_{n_n}$$

Since the power of a noise source is proportional to the square of its RMS value ($P = V_{n_{rms}}^2$), we get:

$$V_{n_{total}}^2 = V_{n_1}^2 + V_{n_2}^2 + \cdots + V_{n_n}^2$$

Taking the square root gives:

$$V_{n_{total}} = \sqrt{V_{n_1}^2 + V_{n_2}^2 + \cdots + V_{n_n}^2}$$

Examples of calculating Noise

Example 1: Thermal Noise and Shot Noise in a Photodetector

Consider a photodetector with two primary noise sources: **thermal noise** from a resistor and **shot noise** from the photodetector current.

1. Thermal Noise ($V_{n_{thermal}}$):

$$v_{n,rms} = \sqrt{4 k T R B}$$

2. Shot Noise ($I_{n_{shot}}$):

$$i_{shot,rms} = \sqrt{2 q I B}$$

If the shot noise current passes through a transimpedance amplifier with a feedback resistor R_f , it contributes to a voltage noise:

$$V_{n_{shot}} = I_{n_{shot}} \times R_f = R_f \times \sqrt{2 q I B}$$

To find the total output noise ($V_{n_{total}}$) of the TIA with both noise components:

$$V_{n_{total}} = \sqrt{V_{n_{thermal}}^2 + V_{n_{shot}}^2}$$

Example: Total Noise in a Photodetector System

Suppose a photodetector system includes:

- **Thermal Noise** from a resistor: $V_{n_{thermal}} = 5 \text{ mV}$
- **Shot Noise** from the photodetector current: $V_{n_{shot}} = 3 \text{ mV}$
- **Amplifier Noise** from the TIA: $V_{n_{amp}} = 2 \text{ mV}$

The total noise RMS value would be:

$$V_{n_{total}} = \sqrt{(5 \text{ mV})^2 + (3 \text{ mV})^2 + (2 \text{ mV})^2}$$

$$V_{n_{total}} = \sqrt{25 + 9 + 4} \text{ mV} = \sqrt{38} \text{ mV} \approx 6.16 \text{ mV}$$

Definition of Bandwidth B

Bandwidth (B) refers to the range of frequencies over which a system, component, or signal operates effectively.

In the context of noise calculations, B is the **frequency range** through which noise is filtered or allowed to pass.

The noise power is directly proportional to the bandwidth: as the bandwidth increases, more noise power passes through, and vice versa.

How do we calculate Δf / B

There are different bandwidths to consider, depending on the context of the system:

Signal Bandwidth

- This is the frequency range over which the actual signal is transmitted or detected.
- If you are analyzing the noise in a communication system, the signal bandwidth is often used as Δf
- **Example:** For a radio signal modulated over a frequency range of 10 kHz, the signal bandwidth is $\Delta f = 10$ kHz

Amplifier Bandwidth

- In systems where an **amplifier** (e.g., a TIA, voltage amplifier, or operational amplifier) is present, the bandwidth of the amplifier limits the range of frequencies over which noise can be amplified.
- The amplifier bandwidth, typically specified as the **-3 dB bandwidth**, is often used as Δf in noise calculations.
- **Example:** If a transimpedance amplifier (TIA) has a -3 dB bandwidth of 1 MHz, the noise calculation should use $\Delta f = 1$ MHz

Filter Bandwidth

- If the signal passes through an **electronic filter**, the bandwidth of that filter determines Δf
- Filters are used to limit noise contributions by restricting the frequency range that can pass through.
- **Example:** A low-pass filter with a cutoff frequency of 100 kHz has a bandwidth of $\Delta f = 100$ kHz

Measurement Instrument Bandwidth

- When using instruments like **oscilloscopes** or **spectrum analyzers**, the measurement device itself has a limited bandwidth, which may affect noise readings.
- The **instrument bandwidth** may be specified and needs to be considered when interpreting noise measurements.
- **Example:** If an oscilloscope has a bandwidth of 20 MHz, then the noise observed on the oscilloscope is limited by $\Delta f = 20$ MHz
 $\Delta f = 20 \text{ MHz}$

Example

Imagine I had a photodetector in reverse bias with light shining on it connected to a 10 kOhm resistor. We measure the voltage across this resistor. A) What are the sources of noise for the voltage measured across the device? B) If I have 1 mA of photocurrent in my photodetector, what will the actual RMS noise level of the voltage measured across the resistor be? Assume a sampling rate of 200 kHz (leading to a bandwidth of 100 KHz).

Given:

- Photocurrent:

$$I = 1 \text{ mA} = 1 \times 10^{-3} \text{ A}$$

- Resistor:

$$R = 10 \text{ k}\Omega = 10^4 \Omega$$

- Sampling rate:

$$f_s = 200 \text{ kHz}$$

⇒ bandwidth (Nyquist):

$$B = 100 \text{ kHz} = 1 \times 10^5 \text{ Hz}$$

- Assume room temperature $T \approx 300 \text{ K}$
- Electron charge $q \approx 1.6 \times 10^{-19} \text{ C}$
- Boltzmann constant $k \approx 1.38 \times 10^{-23} \text{ J/K}$

Example

Imagine I had a photodetector in reverse bias with light shining on it connected to a 10 kOhm resistor. We measure the voltage across this resistor. A) What are the sources of noise for the voltage measured across the device? B) If I have 1 mA of photocurrent in my photodetector, what will the actual RMS noise level of the voltage measured across the resistor be? Assume a sampling rate of 200 kHz (leading to a bandwidth of 100 KHz).

1. Shot noise from the photocurrent

Shot noise **current** RMS over bandwidth B :

$$i_{\text{shot,rms}} = \sqrt{2qIB}$$

Plug in:

$$i_{\text{shot,rms}} = \sqrt{2 \cdot (1.6 \times 10^{-19}) \cdot (1 \times 10^{-3}) \cdot (1 \times 10^5)}$$

Calculate inside the sqrt:

$$2 \cdot 1.6 \times 10^{-19} \cdot 1 \times 10^{-3} \cdot 1 \times 10^5 = 3.2 \times 10^{-17}$$

So:

$$i_{\text{shot,rms}} = \sqrt{3.2 \times 10^{-17}} \approx 5.7 \times 10^{-9} \text{ A}$$

So the shot noise **current** is about:

$$i_{\text{shot,rms}} \approx 5.7 \text{ nA}$$

Convert this to **voltage noise** across the 10 kΩ resistor:

$$v_{\text{shot,rms}} = i_{\text{shot,rms}} \cdot R = (5.7 \times 10^{-9}) \cdot (1 \times 10^4) \approx 5.7 \times 10^{-5} \text{ V}$$

So:

$$v_{\text{shot,rms}} \approx 57 \mu\text{V}$$

Example

Imagine I had a photodetector in reverse bias with light shining on it connected to a 10 kOhm resistor. We measure the voltage across this resistor. A) What are the sources of noise for the voltage measured across the device? B) If I have 1 mA of photocurrent in my photodetector, what will the actual RMS noise level of the voltage measured across the resistor be? Assume a sampling rate of 200 kHz (leading to a bandwidth of 100 KHz).

2. Thermal (Johnson) noise of the 10 kΩ resistor

Thermal noise voltage RMS:

$$v_{th,rms} = \sqrt{4kTRB}$$

Plug in:

$$4kTRB = 4 \cdot (1.38 \times 10^{-23}) \cdot 300 \cdot 10^4 \cdot 10^5$$

First the constants:

- $4 \cdot 1.38 \times 10^{-23} \cdot 300 \approx 1.656 \times 10^{-20}$
- Multiply by $10^4 \cdot 10^5 = 10^9$;

$$4kTRB \approx 1.656 \times 10^{-20} \cdot 10^9 = 1.656 \times 10^{-11}$$

Now take the square root:

$$v_{th,rms} = \sqrt{1.656 \times 10^{-11}} \approx 4.1 \times 10^{-6} \text{ V}$$

So:

$$v_{th,rms} \approx 4 \mu\text{V}$$

Imagine I had a photodetector in reverse bias with light shining on it connected to a 10 kOhm resistor. We measure the voltage across this resistor. A) What are the sources of noise for the voltage measured across the device? B) If I have 1 mA of photocurrent in my photodetector, what will the actual RMS noise level of the voltage measured across the resistor be? Assume a sampling rate of 200 kHz (leading to a bandwidth of 100 KHz).

3. Total RMS noise voltage

The shot and thermal noises are uncorrelated, so total RMS voltage is:

$$\begin{aligned} v_{\text{total,rms}} &= \sqrt{v_{\text{shot,rms}}^2 + v_{\text{th,rms}}^2} \\ &\approx \sqrt{(57 \mu\text{V})^2 + (4 \mu\text{V})^2} \approx 57 \mu\text{V} \end{aligned}$$

(Thermal noise is much smaller, so the total is essentially dominated by shot noise.)

Let's talk about a real world system



In the computer, we can define a signal as an array of numbers.

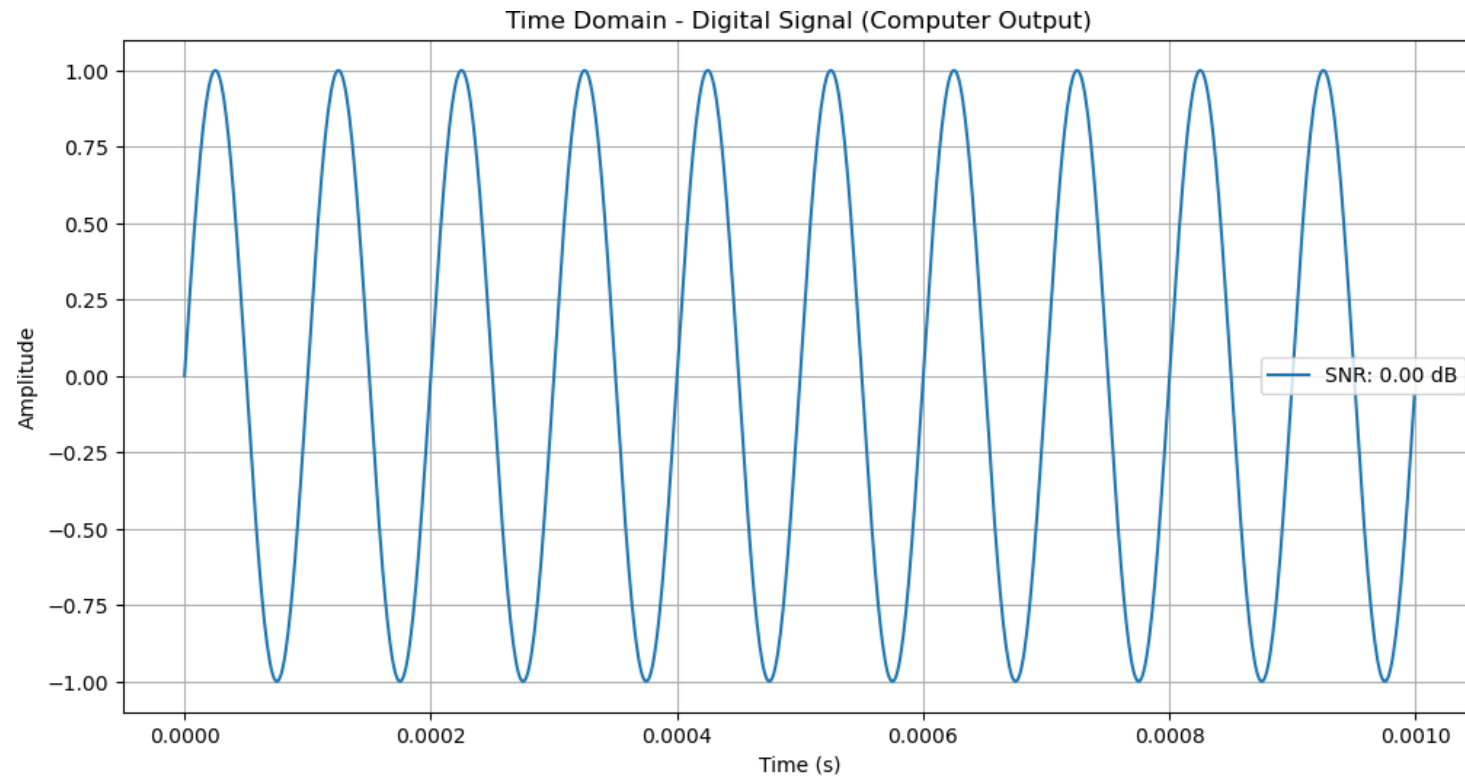
If we wanted to define a sine wave, we could define it as an array of double precision numbers, where each point in the array corresponded to a given time.

Integers from -2^{53} to 2^{53} ($-9,007,199,254,740,992$ to $9,007,199,254,740,992$) can be exactly represented

We could then take this range, which is 2^{54} and spread our output voltage range across that, giving us a very precise number for each time step.

However, there is still going to be one level per timestep, so there is some level of quantization in both voltage and time

Let's talk about a real world system



Let's talk about a real world system



However, we can't drive an led with a double precision number, that's just a bunch of 1's and 0's

We need to convert those into an actual voltage value to apply to the LED.

We would use a digital to analog converter (DAC) to do this. This takes in some number of bits, and depending on the values, it outputs the corresponding analog signal

This DAC needs to be able to take an input, and “instantly” change the output analog voltage to reflect that input.

Let's talk about a real world system



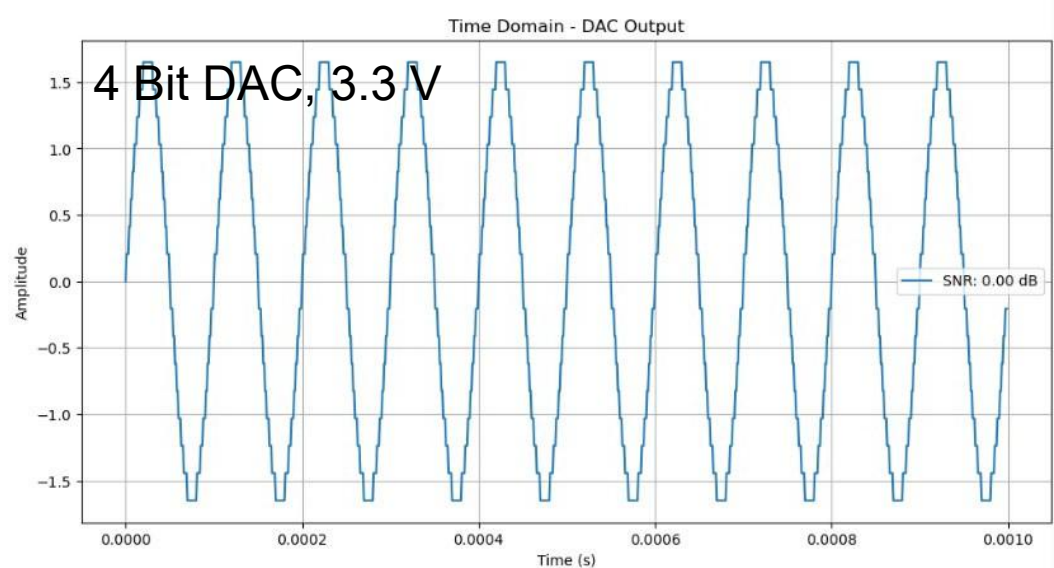
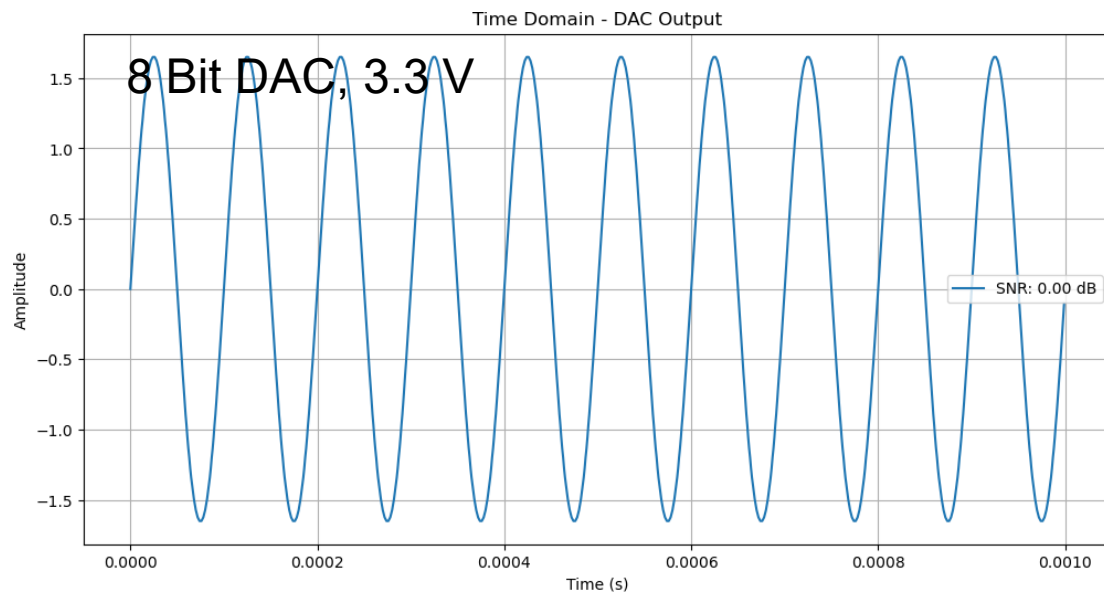
There are a number of noise sources that are relevant:

- **Quantization Noise:** Introduced by the discrete steps used to represent a continuous signal, dependent on DAC resolution.
- **Thermal Noise:** Arises from resistive components, increases with temperature and resistance.
- **Nonlinearity (Distortion Noise):** Due to imperfections in the DAC's response, results in harmonic distortion.
- **Clock Jitter:** Variations in clock timing, affecting precision especially at higher frequencies.
- **Power Supply Noise:** Coupling from power supply fluctuations into the analog output.

Let's talk about a real world system



What is the number of bits for a DAC? It defines the number of different output levels it can generate. A 2 bit DAC would generate 4 values, a 10 bit DAC would generate 1024 values



Let's talk about a real world system



There will be a circuit which drives the LED to the appropriate values, and the LED itself, which will also introduce noise based on the device properties. The key noise components will be:

- **Thermal Noise:**

- Found in the resistive elements of the LED driver circuit.
- It is directly proportional to temperature and resistance.

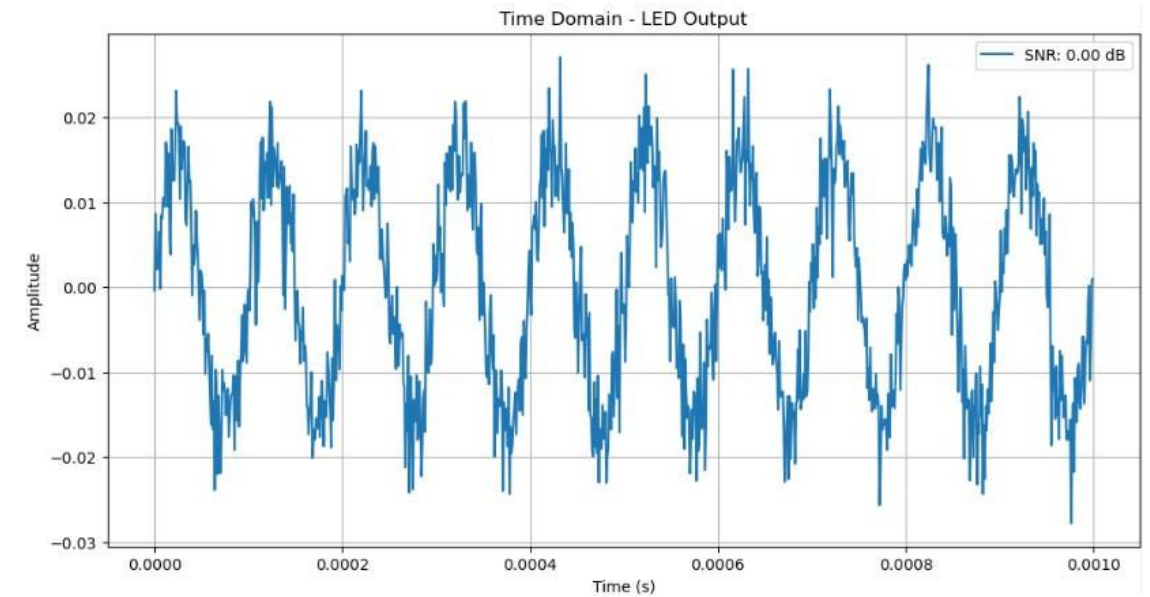
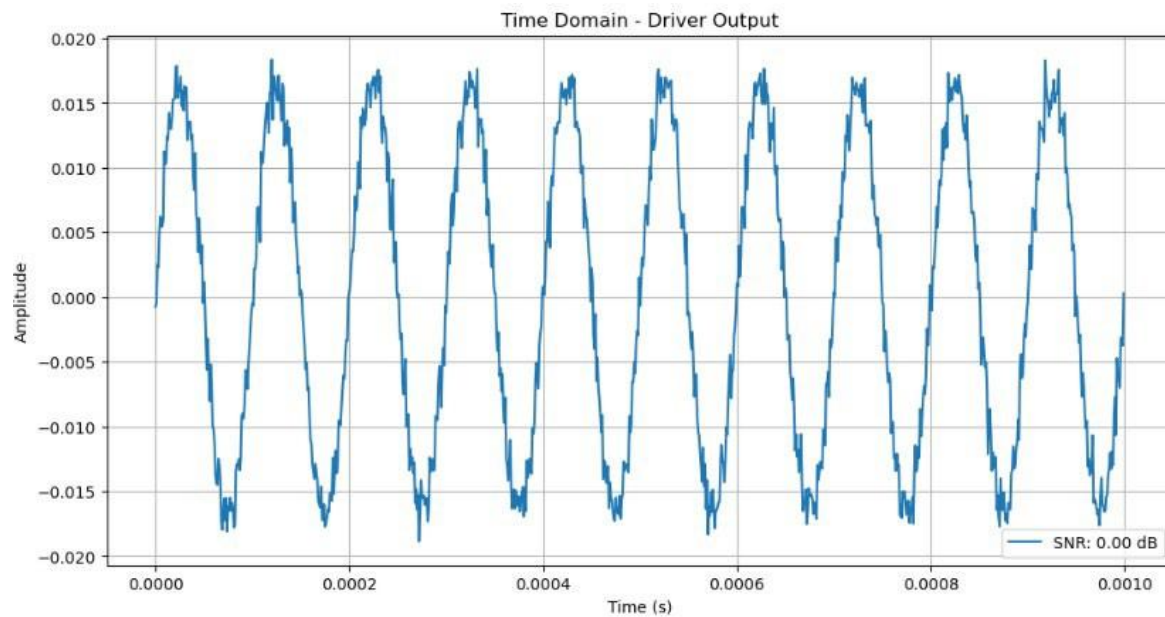
- **Shot Noise:**

- Found in the LED itself due to the random nature of current flow.
- Proportional to the square root of the average current through the LED.

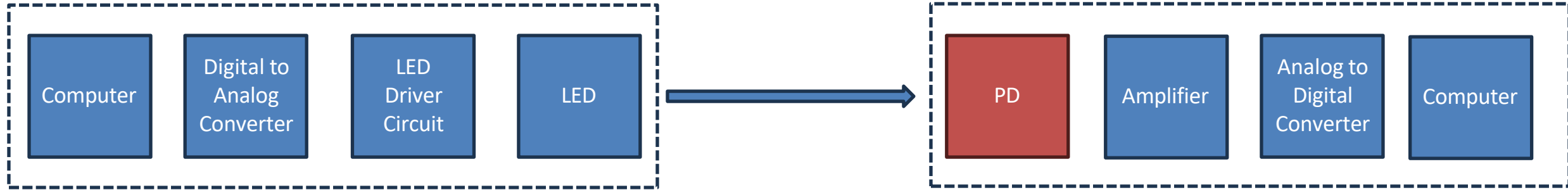
- **Flicker Noise:**

- Found in driver components due to material imperfections.
- Dominates at low frequencies and adds to the overall noise in the driver stage.

Let's talk about a real world system



Let's talk about a real world system



What noise in photodetectors do we see?

- **Shot Noise:**

- Due to the statistical variation in the arrival of photons.
- Significant at low light levels.
- White noise, with uniform power distribution across all frequencies.

- **Thermal Noise:**

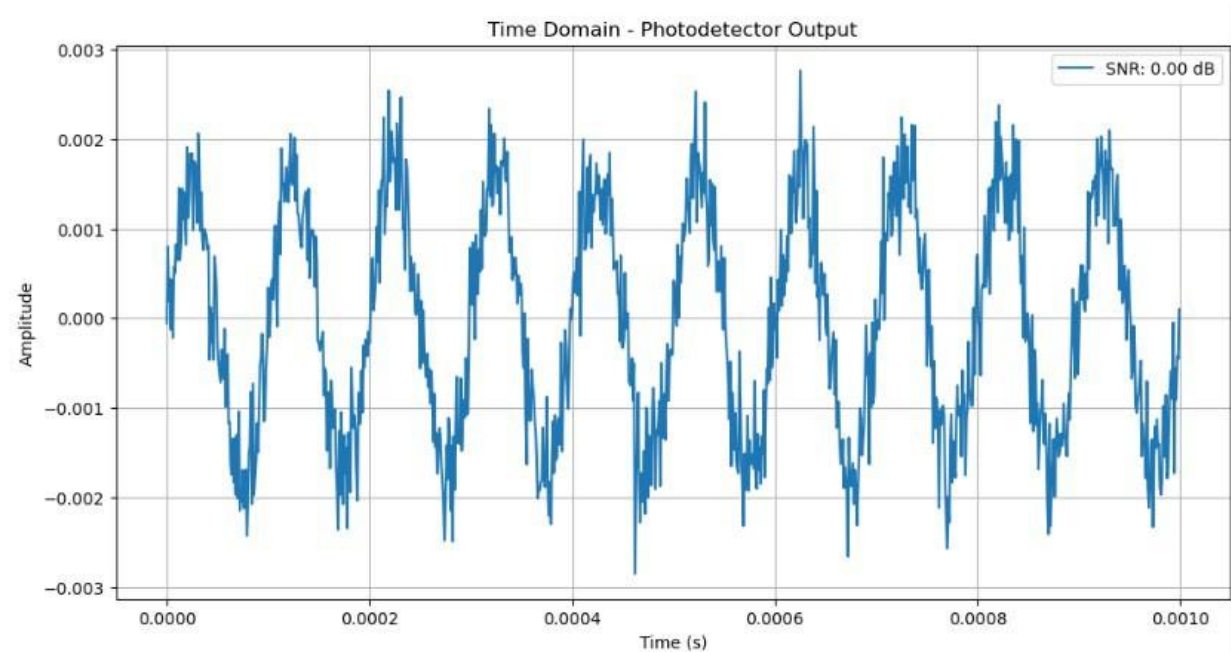
- Due to thermal agitation of electrons in resistive components.
- Increases with temperature and resistance.
- Present in the circuit regardless of light exposure.

- **Dark Current Noise:**

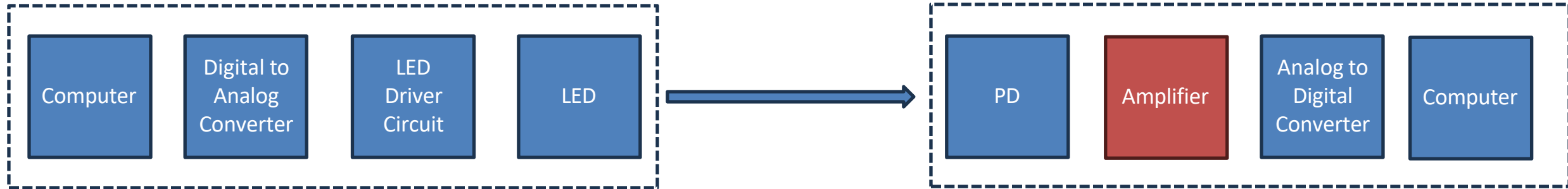
- Due to thermally generated carriers in the absence of light.
- Increases with temperature.
- Reduces signal accuracy, especially in low-light applications.

- **Flicker Noise (1/f Noise):**

- Caused by material imperfections and traps.
- Prominent at low frequencies.
- Inversely proportional to frequency, affecting low-frequency applications.



Let's talk about a real world system



What noise in amplifiers do we see?

- **Thermal Noise from Feedback Resistor:**

- Arises from the feedback resistor.
- Increases with resistance and temperature.
- Affects the output directly as voltage noise.

- **Shot Noise from Photocurrent:**

- Originates from the statistical nature of the current generated by the photodetector.
- Amplified by the TIA, contributing to the output noise.

- **Op-Amp Voltage Noise:**

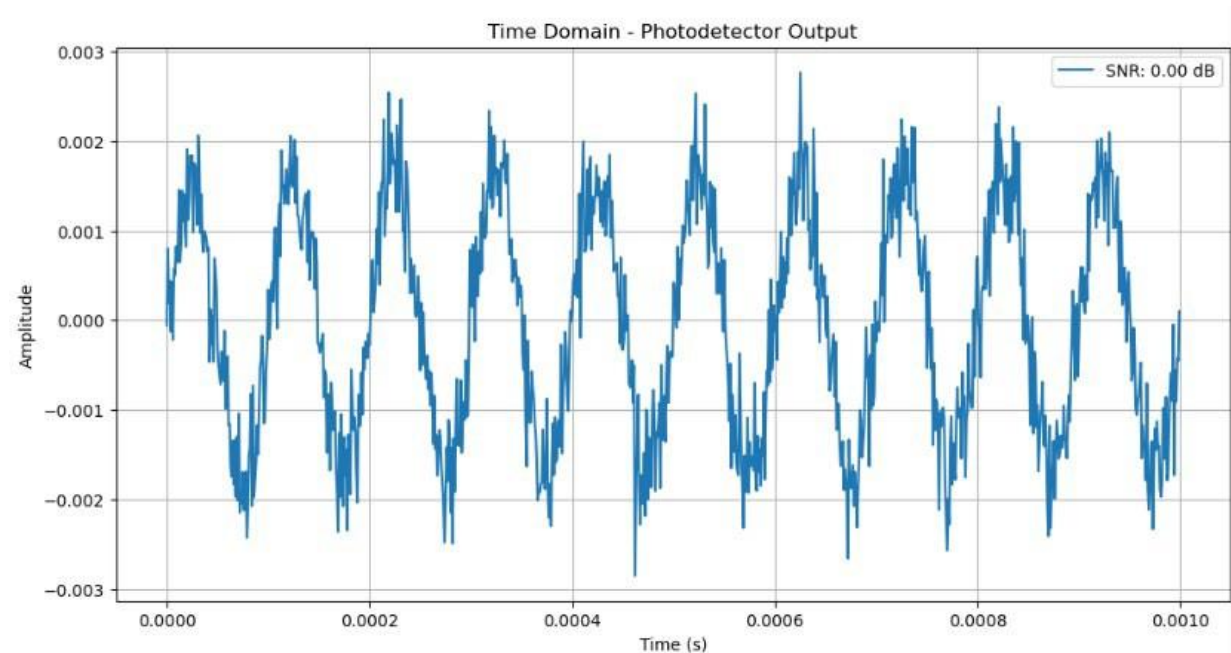
- Intrinsic noise from the operational amplifier.
- Modeled as a noise voltage at the input.
- Amplified by the gain of the TIA.

- **Op-Amp Current Noise:**

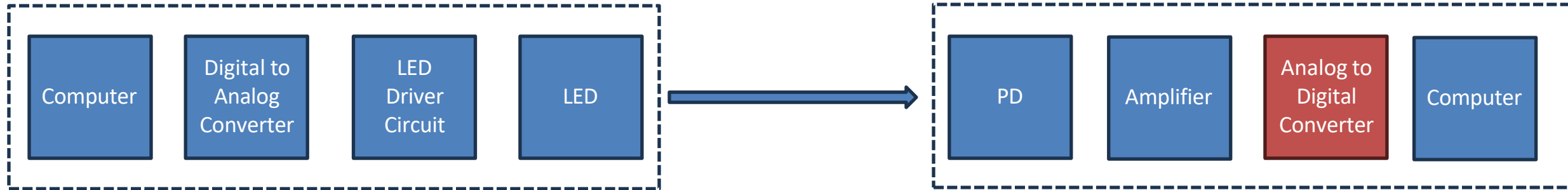
- Due to bias currents of the op-amp transistors.
- Creates additional voltage noise when interacting with the impedance of the circuit.

- **Flicker Noise (1/f Noise):**

- Low-frequency noise due to imperfections in the op-amp.
- Dominant at low frequencies, decreasing with increasing frequency.

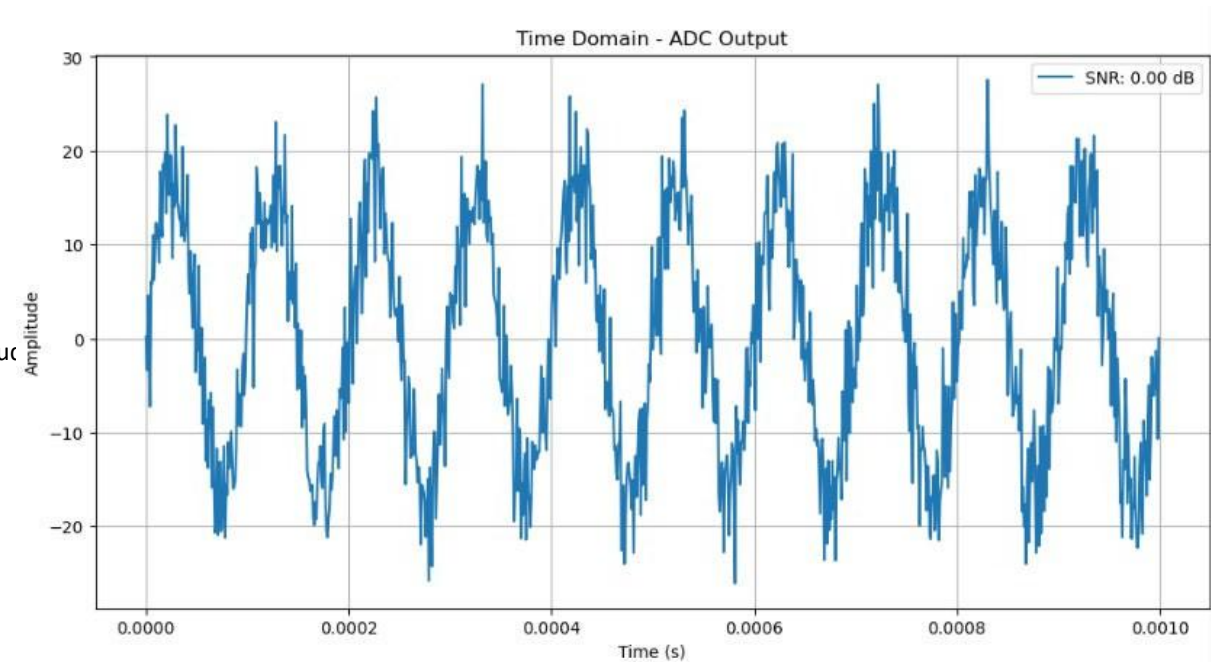


Let's talk about a real world system



What noise do we see in the ADC?

- **Quantization Noise:**
 - **Cause:** Due to approximating a continuous signal with discrete levels.
 - **Characteristics:** Inversely proportional to the number of bits. Modeled as white noise.
- **Thermal Noise:**
 - **Cause:** Due to resistive elements in the analog front end.
 - **Characteristics:** Proportional to temperature and resistance. Sets a noise floor for the ADC.
- **Sampling (Aperture) Jitter:**
 - **Cause:** Timing uncertainty during sampling.
 - **Characteristics:** Affects high-frequency signals more severely, proportional to input frequency and signal amplitude.
- **Reference Voltage Noise:**
 - **Cause:** Noise or fluctuation in the reference voltage.
 - **Impact:** Leads to inaccurate conversions if the reference is not stable.
- **Clock Noise:**
 - **Cause:** Instability or noise in the sampling clock signal.
 - **Impact:** Affects the timing of samples, resulting in inaccuracies.
- **Nonlinearity (Distortion Noise):**
 - **Cause:** Imperfections in the ADC's transfer function.
 - **Impact:** Causes harmonic distortion of the signal, seen as additional unwanted frequency components.
- **Flicker Noise (1/f Noise):**
 - **Cause:** Semiconductor imperfections and low-frequency irregularities.
 - **Characteristics:** Dominates at low frequencies and can degrade signal quality in low-frequency applications.



Let's talk about a real world system

